

Annu Tiwari

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LinkedIn | GitHub | LeetCode | Portfolio

SUMMARY

AI/ML engineer with hands-on experience in signal processing, feature engineering, and classical ML applied to healthcare and brain-computer interface systems. Published IEEE research on low-cost EEG hardware; built full-stack applications end-to-end. Solved 250+ DSA problems on LeetCode.

TECHNICAL SKILLS

Machine Learning: Supervised Learning, Explainable AI, Feature Engineering, Data Preprocessing, Model Evaluation, Scikit-learn, NumPy, Pandas

Programming Languages: Java, Python, JavaScript, SQL

Frameworks & Technologies: Next.js, React, Node.js, Express.js, MongoDB (MERN Stack), REST APIs, Tailwind CSS

Software Tools & Platforms: Git, GitHub, VS Code, Postman, Jupyter Notebook, Google Colab, Overleaf

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

PROJECTS

INORA – Multi-Channel EEG-Based Control

Dec 2025 – Jan 2026

Brain-Computer Interface Project

Technologies: AD620, TL072, ADS1115, ESP32, Python, NumPy

- Built a low-cost 3-channel EEG acquisition system using AD620, ADS1115, and ESP32 with wireless streaming at **83 Hz/channel**.
- Developed a Python pipeline for EEG acquisition, filtering, and DRL-based noise reduction, and validated the system on **20 volunteers**.
- Demonstrated real-time EEG-based Pong game control and recorded physiologically valid EEG signals in the **15–80 μ V** range using hardware costing approximately **INR 3,146**.

BrainScore – EEG-Based Depression Detection

Mar 2025 – Nov 2025

Machine Learning Project

Technologies: Python, Pandas, Scikit-learn, LightGBM, SHAP, Google Colab

- Processed EEG recordings from **58 subjects** using **20-channel** signals and extracted **18** statistical, frequency, and entropy-based biomarkers.
- Built and evaluated six machine learning models, using **SHAP** to interpret feature importance and improve model explainability.
- Achieved **95.28%** classification accuracy with a **LightGBM** model for automated depression detection.

AHAR – Food Donation System

Mar 2024 – Apr 2024

Full Stack Web Application

Technologies: MongoDB, Express.js, React.js, Node.js, Google Maps API

- Developed a food donation platform connecting restaurants, events, societies, and NGOs.
- Implemented real-time tracking, route mapping, and delivery coordination using Google Maps API.
- Enabled transparent food distribution while creating employment opportunities for delivery agents.

EXPERIENCE

AI Engineer Intern

Dec 2025 – Feb 2026

Infosys Springboard

Remote

- Cleaned and preprocessed cricket match datasets and engineered performance-based features.
- Trained machine learning models to predict player runs and wickets for upcoming matches.
- Built an interactive dashboard to visualize predictions and player performance trends.

Machine Learning & Data Science Intern

May 2025 – Jul 2025

Svaarogyam Medical Devices Pvt. Ltd.

Remote

- Worked on healthcare machine learning tasks including data cleaning, preprocessing, feature extraction, and predictive modeling.
- Built end-to-end machine learning pipelines for healthcare use cases.
- Collaborated with a remote team to deliver project milestones in the medical devices domain.

EDUCATION

IILM University, Greater Noida

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 8.74/10

CERTIFICATIONS

- Research Paper accepted at IEEE Sponsored ETNCC 2026 for “INORA: A Low-Cost 3-Channel EEG Acquisition System with Driven-Right-Leg Noise Suppression for BCI Prototyping”
- Hackathon Winner (1st Rank) – Secured 1st place among 1,004 participants (118 teams)
- Secured 1st place in the Call for Paper event at SRM Institute of Science and Technology
- Completed Infosys Springboard Pragati: Path to Future – Cohort 6
- Completed Data Structures & Algorithms (Sigma Batch) from Apna College
- Solved 250+ Data Structures & Algorithms problems on LeetCode